

Natural Disasters Risk: A Multi-Source Dashboard

Interactive Visualization — Assignment 2 (Concept & Dashboard) · Krenar · JKU Linz

Topic

Disaster risk reduction agencies need to anticipate where humanitarian crises will overwhelm national response capacity. The INFORM Risk Index operationalises this with a composite score combining hazard exposure, population vulnerability, and lack of coping capacity. This dashboard lets a user explore the global INFORM risk landscape and, crucially, compare predicted risk against the disaster losses actually observed in EM-DAT over the last three decades — surfacing both well-modelled countries and outliers that the index over- or under-estimates.

Dataset

INFORM Risk Index (Joint Research Centre, European Commission) — open-source composite risk assessment covering 191 countries, scored 0–10 across three dimensions (Hazard & Exposure, Vulnerability, Lack of Coping Capacity) and ~80 underlying indicators. Hazard-specific sub-scores are available for flood, earthquake, tropical cyclone, drought, tsunami, and epidemic. Five years of comparable trend data (2022–2026). Licensed CC-BY-4.0; downloaded as a single Excel workbook from drmkc.jrc.ec.europa.eu/inform-index.

EM-DAT International Disaster Database (CRED, UCLouvain) — event-level records of 27,000+ disasters from 1900 to present, with deaths, persons affected, and economic damages per event. We use natural-hazard events from 1995 onward to give a 30-year window of comparable modern records. Free for academic use; downloaded as Excel after free registration at emdat.be.

Preprocessing. A Python script (*preprocess.py*, included) loads both files, normalises country names to ISO3 codes, filters EM-DAT to natural hazards, and joins the two on country code. Outputs are four tidy CSVs imported into Power BI: *inform_latest*, *inform_trend*, *emdat_events*, and *joined_summary*.

User Tasks

- **T1 — Identify:** which countries face the highest composite risk, and which hazard types dominate their profile.
- **T2 — Decompose:** for a selected country, break down the INFORM score into its three dimensions and read the indicator-level drivers.
- **T3 — Compare:** place a country against its regional peers and against its own five-year trajectory.
- **T4 — Reconcile:** contrast predicted INFORM risk with observed EM-DAT losses (deaths, affected, damages) to surface under- and over-predicted countries.

Dashboard (Power BI, 4 pages)

- **Global Overview** — choropleth of INFORM Risk; top/bottom-10 ranking; hazard-type filter; year slider. Drives cross-page filtering.
- **Country Profile** — selected country's three-dimension breakdown, hazard-specific scores, 5-year trend line, and regional benchmark.
- **Event History** — EM-DAT timeline (stacked area by hazard type), with parallel views for deaths, persons affected, and damages.
- **Risk vs. Reality** — scatter of INFORM score against 30-year loss metric, log y-axis, region colour-coded; the diagonal residual flags outliers and motivates Assignment 3.

Outlook (Assignment 3)

The reconciliation view (page 4) is the obvious anchor for a custom D3 component: a radial “risk fingerprint” per country — petals encoding the hazard-specific scores, brightness encoding realised losses — with a linked event scatter. This pushes beyond Power BI's stock charts on both the visual encoding (custom radial geometry, dual encoding) and the interaction (brushing across linked views).